

Intertemporal Optimization of Formation Control for Nonlinear Networks with Multiple Constraints: A Reach-Avoid Game Approach

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Abstract—This paper, by using viability theory, presents a novel coordinate-free and optimal formation control protocol for nonlinear network systems with control, communication, connection and collision avoidance constraints. Existing numerical tools of viability theory characterize the feasible region of states in which a solution exists with respect to two-player reach-avoid games for constrained complex systems. This work formulates the optimal formation control problem as a multi-player reach-avoid differential graphical game based on viability theory, ensuring the formation control law adapts to the fastest convergence rate for the switching local networks while satisfying all the constraints by employing local information. The network is assumed to be always unknown and switching, where the edges of its graph are allowed to be temporarily disconnected while the connectivity of graph is guaranteed. The value function is approximated by the adaptive graph neural network (GNN), where its parametric domains are characterized by viability theory. Besides, the convergence of the approximation errors of the value function for the differential graphical game is analyzed. The effectiveness of the proposed method is confirmed and illustrated via simulations.

Index Terms—Formation control, multi-agent system, viability theory, differential graphical games, collision avoidance, connectivity maintenance, constrained networks.

I. INTRODUCTION

The formation control technique has been extensively applied to various areas involving multi-agent network systems, such as satellite networks and autonomous ocean sampling networks [1], [2]. While consensus control algorithms aim to achieve agreement on the states of all agents in dynamic

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network systems, formation control algorithms further refine this by positioning agents based on their relative states [3]. Fax and Murray established the information flow law in the formation control to stabilize the location structure of multiple vehicles [4], [5], and later connected graph Laplacian and consensus control principles with formation control problems [6].

This paper investigates multi-pursuer and multi-evader reach-avoid differential graphical games in the context of formation control for large-scale nonlinear switching networks with multiple constraints, using viability theory based on local information. Viability theory characterizes the reachability and viability of states via capture basins and viability kernels, respectively. Existing numerical tools from viability theory can generate corresponding control strategies [7], compute reach-avoid sets, and determine value functions for differential games [7], [8]. These have been widely applied in autonomous racing, navigation, obstacle avoidance, and safety management of high-speed autonomous vehicles [9]–[13]. Existing work on reach-avoid optimal control for constrained nonlinear systems focus on two-player reach-avoid differential games in the context of viability theory. In multi-player reach-avoid games, most studies consider multiple pursuers and a single evader [14]–[17], with few addressing multi-pursuer and multi-evader scenarios [18].

Viability theory assists formulating the optimal formation control problem as a reach-avoid differential graphical game. The main challenges include: formulating the optimal formation control problem for network systems with multiple constraints as a multi-player reach-avoid differential graphical game; finding and proving the Hamilton-Jacobi-Bellman (HJB) equation, for which the value function is the viscosity solution in multi-player reach-avoid differential graphical games; and approximating the value function by solving the HJB equation and proving the convergence of the Bellman error. The main motivations in this paper are threefold: ensuring all agents to reach the desired formation within finite time while satisfying control, communication, connection and collision avoidance constraints; maintaining network connectivity under communication and connection constraints despite unknown and switching graph topology; and enabling on-line optimization of the formation control strategy using only local information, with each agent's control strategy converging to a Nash equilibrium.

Compared with the existing works, the main innovations

and contributions of the proposed work are as follow:

- 1) Formulation of optimal formation control problem: the optimal formation control problem for networks with multiple constraints is formulated as a multi-player reach-avoid game without constraints using viability theory. The constraints include control, communication, connection, and collision avoidance, unlike most existing works which consider no more than two constraints [19]–[21].
- 2) Design of control law: the designed control law enables the reach-avoid differential graphical game to achieve a Nash equilibrium with guaranteed connectivity maintenance, state convergence, and collision avoidance. Most existing works address these issues in formation control without optimization [22]–[24].
- 3) Graph topology: the proposed approach allows the graph to be unknown and switching, where graph edges can be temporarily disconnected while maintaining overall connectivity. This contrasts with existing works which solve the connectivity maintenance problems using artificial potential functions that do not allow edge disconnections once connected [22], [25].

The rest of this paper is organized as follows. Section II introduces the configuration of the network system and preliminary knowledge of viability theory. Section III formulates the formation control problems as reach-avoid differential graphical games, and discusses the formation control framework. Section IV illustrates the effectiveness of the proposed algorithm. Section V concludes this paper.

II. PROBLEM FORMULATION

This section clarifies the basic configurations of the optimal formation control problem, and explains how to formulate the formation control problem as a reach-avoid differential graphical game.

A. Network System

Throughout this paper, the network contains N agents, and the i th agent is denoted as v_i , i.e., the i th node of its graph $\mathcal{G}$. The set of indices of the neighbour agents connected to v_i is denoted as $\mathcal{N}_i$. The states of the i th agent v_i in the global coordinate frame are denoted as $e_i \in \mathbb{R}^n$ including its position $[x_i, y_i]^T$ and orientation $\theta_i \in [-\pi, \pi]$ (only 2-D space is considered in this work, and while 3-D space can be addressed similarly).

The network system can be partitioned to pairwise systems of connected agents. As only local information can be used, by denoting $\mathcal{R}(\theta_i)$ as the clockwise rotation matrix, the relative state e_{ij} between v_i and v_j can be written as $e_{ij} = \mathcal{R}(\theta_i)(e_j - e_i) \in \mathbb{R}^n$ in the local coordinate frame of v_i , where $\theta_{ij} = \theta_j - \theta_i$. The desired local formation framework is constructed by the desired relative state $h_{ij} = \mathcal{R}(\theta_i)(h_j - h_i)$ in the local coordinate frame of v_i , where h_j and h_i are the desired states of v_i and v_j in the global formation framework, respectively. Then, $e_{ij}^h = h_{ij} - e_{ij}$ is the local formation relative state. Those pairwise agents have the following properties:

- 1) The pair of agents v_i and v_j can only communicate within the communication zone $\mathbb{K}$ of e_{ij} , i.e., $e_{ij} \in \mathbb{K}$.

- 2) Only collision avoidance among agents is considered in terms of the safety of the network system. Denoting compact sets $\mathbb{B} \subset \mathbb{K}$ as the collision set of e_{ij} , a collision occurs between v_i and v_j if $e_{ij} \in \mathbb{B}$.
- 3) Denoting $\mathbb{C}$ as the target set of e_{ij}^h enclosing the original point which is designed depending on the allowable convergence offset of e_{ij}^h , the formation of the network is achieved if $e_{ij}^h \in \mathbb{C}$, for $\forall i \in [1, N]$, $j \in \mathcal{N}_i$.

The graph of the network is configured as shown below:

- 1) The desired relative state h_{ij} and the relative state e_{ij} including θ_{ij} in the local coordinate frame of v_i are always given for $\forall i \in [1, N]$, $\forall j \in \mathcal{N}_i$.
- 2) The connectivity of the graph is subject to both communication constraint and the reachability of the agents.
- 3) The graph is assumed to be undirected, unknown and switching, where the weights on its edges are unified as $a_{i,j}$, and $a_{i,j} > 0$ if and only if the nodes v_i and v_j are connected, otherwise $a_{i,j} = 0$.

The aims of the designed formation control law can be summarized as follows:

- 1) The formation can be achieved in a finite time T (i.e. $\forall i \in [1, N]$, $j \in \mathcal{N}_i$, $e_{ij}^h(T) \in \mathbb{C}$), and is retained after T (i.e. once $e_{ij}^h(T) \in \mathbb{C}$, then $\forall t \in [T, \infty]$, $e_{ij}^h(t) \in \mathbb{C}$).
- 2) The network is always connected, and satisfies all the constraints for $\forall t \in [0, \infty]$: for $\forall i \in [1, N]$, $u_i(t) \in U$; for $\forall e_{ij}(t) \in \mathbb{K}$, $\forall e_{ij}(t) \in \mathbb{B}^c$; for $\forall j \in \mathcal{N}_i$, $\exists u_i(t), u_j(t) \in U$, $e_{ij}(t) \in \mathbb{K}$, $\exists \tau \leq 0$, $e_{ij}(t + \tau) \in \mathbb{C}$.
- 3) The optimal distributed control law is always adapting to the fastest convergence rate of e_{ij}^h for different local networks while satisfying all the constraints and connectivity maintenance.

B. System Reformulation

The agent v_i considered in this work is an autonomous vehicle with the following dynamics:

$$\dot{e}_i = \mathbf{f}(e_i) + \mathbf{g}(e_i)u_i, \quad (1)$$

where $u_i \in U$, and U stands for the convex control constraint, and the system dynamics of pairwise agents, for the local coordinate frame of v_i , can be written as

$$\begin{aligned} \dot{e}_{ij} &= \mathcal{R}(\theta_i)(\dot{e}_j - \dot{e}_i) + \dot{\mathcal{R}}(\theta_i)(e_j - e_i) \\ &= f(e_{ij}) + g_1(e_{ij})u_i + g_2(e_{ij})u_j, \end{aligned} \quad (2)$$

under the following assumption:

Assumption 1: The dynamics of all the agents are controllable and identical in the network; the functions $\mathbf{f}$ and $\mathbf{g}$ in (1) are uniformly continuous and differentiable; $\mathcal{R}(\theta_i)(\mathbf{f}(e_j) - \mathbf{f}(e_i))$ and $\mathcal{R}(\theta_i)(\mathbf{g}(e_j)u_j - \mathbf{g}(e_i)u_i)$ can be rewritten as functions of $e_{i,j}$.

As the states of only pairwise agents are considered in the formulation (i.e. e_{ij} and e_{ij}^h), each pair of agents share one pairwise system. Thus, the local coordinate frames of agents v_i and v_j have to be unified ensuring that their relative states in the pairwise system are always identical referencing either v_i or v_j . When the local coordinate of v_i is transformed to v_j , then $e_{ji} = -\mathcal{R}(\theta_{ij})e_{ij}$. The local coordinate frames for all

the pairs of connected nodes (v_i and v_j) can be unified by the following transform:

$$e_{i,j} = \begin{cases} e_{ij}, & \text{for } \theta_{ij} \geq 0, \\ -\mathcal{R}(\theta_{ij})(e_{ij}), & \text{for } \theta_{ij} < 0, \end{cases} \quad (3a)$$

$$(3b)$$

and the transformed relative state system $F(e_{i,j})$ is formulated as follows:

$$\dot{e}_{i,j} = f(e_{i,j}) + g_+(e_{i,j})u_i + g_-(e_{i,j})u_j, \quad (4)$$

where

$$g_+(e_{i,j}) = \begin{cases} g_1(e_{i,j}), & \text{for } \theta_{ij} \geq 0, \\ g_2(e_{i,j}), & \text{for } \theta_{ij} < 0, \end{cases} \quad (5a)$$

$$(5b)$$

$$g_-(e_{i,j}) = \begin{cases} g_2(e_{i,j}), & \text{for } \theta_{ij} \geq 0, \\ g_1(e_{i,j}), & \text{for } \theta_{ij} < 0. \end{cases} \quad (5c)$$

$$(5d)$$

The local formation relative state denoted by $e_{ij}^h = h_{ij} - e_{ij}$ applies to (3) and (4) similarly, i.e. $e_{i,j}^h$.

C. Controllability Characterization

The connectivity of the graph and the safety of the network are determined by the capture basin and the viability kernel, respectively, as they characterize the controllability of the network system. The capture basin $\mathbb{R}_{i,j}$ and the viability kernel $\mathbb{V}_{i,j}$ are defined as follows:

Definition 1: The capture basin $\mathbb{R}_{i,j}$ is defined as the subset of all initial states $e_{i,j}^h$, such that $\exists u_i, u_j \in U$ their trajectories reach the target $\mathbb{C}$ in a finite time τ , while $e_{i,j} \in \mathbb{K}$ guarantees the communication between v_i and v_j before reaching the target [7]. It is specified as

$$\mathbb{R}_{i,j} = \left\{ e_{i,j}^h(0) \mid \exists \tau \geq 0, \forall t \in [0, \tau], \right. \\ \left. \exists u_i(t), u_j(t) \in U, e_{i,j}(t) \in \mathbb{K} \wedge e_{i,j}^h(\tau) \in \mathbb{C} \right\}. \quad (6)$$

If and only if $e_{i,j}^h \in \mathbb{R}_{i,j}$, there exists at least one control strategy such that their communication and reachability are guaranteed. The pair of the nodes v_i and v_j are connected when $e_{i,j}^h \in \mathbb{R}_{i,j}$, otherwise they are disconnected.

Remark 1: The target set $\mathbb{C}$ is designed to satisfy $\mathbb{C} \subset \mathbb{R}_{i,j}$, and this implies that $\forall e_{i,j}^h(t) \in \partial \mathbb{C} \subset \mathbb{R}_{i,j}$, $\forall \tau \geq 0$, $\exists u_i(t), u_j(t) \in U, e_{i,j}^h(t + \tau) \in \mathbb{C}$, i.e., once $e_{i,j}^h$ enters $\mathbb{C}$, there exists a control strategy such that $e_{i,j}^h$ can permanently remain inside $\mathbb{C}$.

Definition 2: The viability kernel $\mathbb{V}_{i,j}$ is defined as the set of all initial states $e_{i,j}$ outside $\mathbb{B}$ such that $\exists u_i, u_j \in U$ their trajectories permanently remain outside $\mathbb{B}$ for either $|\mathcal{N}_i| \leq 2$ or $|\mathcal{N}_i| > 2$, and moreover at least one of the trajectories of their transformed state $e_{i,j}^h$ permanently remain inside $\mathbb{R}_{i,j}$ for $|\mathcal{N}_i| \leq 2$ [7]. It is specified as:

$$\mathbb{V}_{i,j} = \left\{ e_{i,j}(0) \mid \forall t \geq 0, \exists u_i(t), u_j(t) \in U, e_{i,j}(t) \in \mathbb{B}^c, \right. \\ \left. \exists u_i(t), u_j(t) \in U, e_{i,j}^h(t) \in \mathbb{R}_{i,j} \wedge e_{i,j}(t) \in \mathbb{B}^c \right\}, \quad (7a)$$

when $|\mathcal{N}_i| \leq 2$, and

$$\mathbb{V}_{i,j} = \left\{ e_{i,j}(0) \mid \forall t \geq 0, \exists u_i(t), u_j(t) \in U, e_{i,j}(t) \in \mathbb{B}^c \right\}, \quad (7b)$$

when $|\mathcal{N}_i| > 2$. The pair of agents v_i and v_j are safe when $e_{i,j} \in \mathbb{V}_{i,j}$, otherwise a collision must occur in a finite time.

If and only if $e_{i,j}^h \in \mathbb{R}_{i,j}$, there exists at least one control strategy such that any potential collisions between v_i and v_j can be avoided. Moreover, the local connectivity of the graph can be determined by $\mathbb{V}_{i,j}$, as it requires $e_{i,j}^h \in \mathbb{R}_{i,j}$ when $|\mathcal{N}_i| \leq 2$. If and only if $i \in [1, N]$, $j \in \mathcal{N}_i$, $e_{i,j} \in \mathbb{V}_{i,j}$, the network is connected and safe, otherwise the network is either disconnected or unsafe.

Remark 2: The configuration of the viability kernel $\mathbb{V}_{i,j}$ allows the graph of the network to be always switching, and v_i and v_j can be disconnected (i.e. $e_{i,j}^h \notin \mathbb{R}_{i,j}$) when $|\mathcal{N}_i| \geq 2$.

The capture basin and the viability kernel are employed to formulate the reach-avoid differential graphical game by relating the arrival time and the maximal exit time functions, respectively. The arrival time function $\varpi(e_{i,j}^h, u_i, u_j)$ is defined as the time for $e_{i,j}^h$ to reach target $\mathbb{C}$, which characterizes the capture basin as follows: $\exists u_i, u_j \in U, \varpi(e_{i,j}^h, u_i, u_j) \in [0, \infty) \Leftrightarrow e_{i,j}^h \in \mathbb{R}_{i,j}$ and $\forall u_i, u_j \in U, \varpi(e_{i,j}^h, u_i, u_j) = \infty \Leftrightarrow e_{i,j}^h \notin \mathbb{R}_{i,j}$.

The maximal exit time function $\gamma(e_{i,j})$ is defined as: when $|\mathcal{N}_i| \leq 2$,

$$\gamma(e_{i,j}(0)) = \sup_{u_i, u_j \in U} \left\{ \tau \geq 0 \mid e_{i,j}^h(0) \in \mathbb{R}_{i,j}, e_{i,j}(0) \in \mathbb{B}^c, \right. \\ \left. e_{i,j}^h(\tau) \in \partial \mathbb{R}_{i,j}^c \vee e_{i,j}(\tau) \in \partial \mathbb{B} \right\}, \quad (8a)$$

i.e., the maximal time for either $e_{i,j}^h$ to leave $\mathbb{R}_{i,j}$ or $e_{i,j}$ to enter $\mathbb{B}$, and when $|\mathcal{N}_i| > 2$,

$$\gamma(e_{i,j}(0)) = \sup_{u_i, u_j \in U} \left\{ \tau \geq 0 \mid e_{i,j}(0) \in \mathbb{B}^c, \right. \\ \left. e_{i,j}(0) \in \mathbb{K}, e_{i,j}(\tau) \in \partial \mathbb{B} \right\}, \quad (8b)$$

i.e., the maximal time for $e_{i,j}$ to enter $\mathbb{B}$, which characterizes the viability kernel as follows: $\exists u_i, u_j \in U, \gamma(e_{i,j}) = \infty \Leftrightarrow e_{i,j} \in \mathbb{V}_{i,j}$ and $\forall u_i, u_j \in U, \gamma(e_{i,j}) \in [0, \infty) \Leftrightarrow e_{i,j} \notin \mathbb{V}_{i,j}$, and then the maximal finite exit time T_λ can be defined as:

$$T_\gamma = \max_{e_{i,j} \in \mathbb{K} \setminus \mathbb{B}} \left\{ \gamma(e_{i,j}) \in [0, \infty) \right\}, \quad (9)$$

where $\gamma(e_{i,j}) = T_\gamma \Leftrightarrow e_{i,j} \in \partial \mathbb{V}_{i,j}^c$. Then, we obtain $\forall e_{i,j}^h \in \mathbb{R}_{i,j}$, $\varpi(e_{i,j}^h, u_i, u_j) \in [0, \infty)$, and $\forall e_{i,j} \in \partial \mathbb{V}_{i,j}$, $\gamma(e_{i,j}) = \infty$. See [26] for the detailed steps to compute the capture basin $\mathbb{R}_{i,j}$ and the viability kernel $\mathbb{V}_{i,j}$.

D. Multi-player Reach-Avoid Game

Based on viability theory, the optimal formation control problem for network systems with multiple constraints is formulated as a multi-player reach-avoid differential graphical game through the arrival time function and the maximal exit time function. The network system for the multi-player game is assumed to satisfy the following assumption:

Assumption 2: The network system is initially connected without any potential collision (i.e. $\forall i \in [1, N], \forall j \in \mathcal{N}_i, e_{i,j}(0) \in \mathbb{V}_{i,j}$); there exists at least one control strategy $u = [u_1, \dots, u_N]$ and a terminal time T such that the formation can be achieved within T with guaranteed connectivity and collision avoidance.

Note that if the network system does not satisfy Assumption 2, there exists no solution to the formation control problem. By denoting the weight on the edge between v_i and v_j as $a_{i,j} > 0$ and $E = [e^T, e^{hT}]^T \in \mathbb{R}^{2nN}$, the designed control law aims to minimize the intertemporal cost function

$$J_i(E(t), u_{\mathcal{N}_i^+}) = \max \left\{ \max_{t \in [t, t+\tau]} \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}(t)) - T_\gamma)^{-1} \right\}, \right. \\ \left. \sum_{j \in \mathcal{N}_i} a_{i,j} \varpi(e_{i,j}^h(t), u_i, u_j) \right\}, \quad (10)$$

where $\tau \in [0, T-t]$ stands for the time that local network $v_{\mathcal{N}_i^+}$ persists before the local graph topology switches. Multi-player reach-avoid differential graphical game has the following properties:

- 1) If and only if the control strategy u for all agents enables the network to reach the desired formation in a finite time, then $\forall t \in [0, T]$, $\forall i \in [1, N]$, $\forall j \in \mathcal{N}_i$, $\sum_{j \in \mathcal{N}_i} a_{i,j} \varpi(e_{i,j}^h(t), u_i, u_j) \in [0, \infty)$ and $\sum_{j \in \mathcal{N}_i} a_{i,j} \varpi(e_{i,j}^h(T), u_i, u_j) = 0$, otherwise, $\exists t \in [0, T]$, $\exists i \in [1, N]$, $\exists j \in \mathcal{N}_i$, $\sum_{j \in \mathcal{N}_i} a_{i,j} \varpi(e_{i,j}^h(t), u_i, u_j) = \infty$.
- 2) If and only if the network satisfies connectivity maintenance and all the constraints, then $\forall t \in [0, T]$, $\forall i \in [1, N]$, $\forall j \in \mathcal{N}_i$, $(\gamma(e_{i,j}(t)) - T_\gamma)^{-1} = 0$, otherwise $\exists t \in [0, T]$, $\exists i \in [1, N]$, $\exists j \in \mathcal{N}_i$, $\max_{t \in [0, T]} \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}(t)) - T_\gamma)^{-1} \right\} = \infty$.
- 3) The cost function in (10) is constructed by picking up the maximum value of two terms such that the prior tasks of the agents focus on satisfying connectivity maintenance and all the constraints, i.e., $(\gamma(e_{i,j}(t)) - T_\gamma)^{-1} = 0$, for $t \in [0, T]$, $\forall i \in [1, N]$, $\forall j \in \mathcal{N}_i$, and then the task of the agents focus on minimizing the time $\sum_{j \in \mathcal{N}_i} a_{i,j} \varpi(e_{i,j}^h(t), u_i, u_j)$ to achieve formation. If the network either dissatisfies one of the constraints or is disconnected, then $J_i = \infty$.

The above properties imply that if and only if the network reaches the desired formation in a finite time while satisfying connectivity maintenance and all the constraints, then $J_i(E(t), u_{\mathcal{N}_i^+}) \in [0, \infty)$ and $J_i(E(T), u_{\mathcal{N}_i^+}) = 0$, otherwise $\exists t \in [0, T]$, $\exists i \in [1, N]$, $J_i(E(t), u_{\mathcal{N}_i^+}) = \infty$. This can be explained by the case of the reach-avoid game between v_i and v_j , where the capture basin and the viability kernel relate $\mathbb{R}_{i,j}$ and $\mathbb{V}_{i,j}$ for a pair of agents v_i and v_j , by the following lemma:

Lemma 1: Let T be any terminal time slot and $e_{i,j}(0) \in \mathbb{V}_{i,j}$, if there exist control strategies u_i and u_j satisfying

$$\max \left\{ \varpi(e_{i,j}^h(t), u_i, u_j), \right. \\ \left. \max_{t \in [0, T]} \left\{ (\gamma(e_{i,j}(t)) - T_\gamma)^{-1} \right\} \right\} \in [0, \infty), \quad (11)$$

then, $e_{i,j}^h(t) \in \mathbb{R}_{i,j}$, $e_{i,j}(t) \in \mathbb{V}_{i,j}$, and $\exists t \geq 0, e_{i,j}^h(t+t) \in \mathbb{C}$ and vice versa.

Proof: See the Appendix. ■

In the multi-player reach-avoid graphical game, when $e_{i,j} \in \hat{\mathbb{V}}_{i,j} \setminus \partial \hat{\mathbb{V}}_{i,j}$, the pairwise system in (2) acts as a pursuer aiming

to achieve the formation (i.e. $J_i(E(T), u_{\mathcal{N}_i^+}) = 0$), and when $e_{i,j} \in \partial \hat{\mathbb{V}}_{i,j}$, the pairwise system acts as an evader aiming to guarantee collision avoidance and connectivity maintenance (i.e. $\forall t \in [0, T]$, $J_i(E(t), u_{\mathcal{N}_i^+}) \in [0, \infty)$). The optimal distributed control strategy u_i^* is designed for all pursuers and evaders to enable the differential graphical game to achieve a Nash equilibrium with guaranteed reachability, connectivity and safety, where the Nash is defined as:

Definition 3: A set of controllers $u_{\mathcal{N}_i^+}^* = \{u_i^*\} : i \in \mathcal{N}_i^+$ constitute a Nash equilibrium, if and only if the intertemporal cost J_i satisfies

$$J_i(E(t), u_{\mathcal{N}_i^+}^*) \leq J_i(E(t), u_i, u_{\mathcal{N}_i}^*), \quad (12)$$

for $\forall i \in [1, N]$, $\forall u_i \in U$, over $[t, t+\tau]$, where the local network $v_{\mathcal{N}_i^+}$ persists for τ seconds before the local graph topology switches.

Assumption 2 guarantee the existence of the solution such that the terminal condition of the corresponding value function

$$V_i(E(t)) = J_i(E(t), u_{\mathcal{N}_i^+}^*) \\ = \inf_{u_i \in U} \{J_i(E(t), u_i, u_{\mathcal{N}_i}^*)\} \quad (13)$$

satisfies $\exists T \in [0, \infty)$, $\forall t \in [0, T]$, $V_i(E(t)) \in [0, \infty)$ and $V_i(E(T)) = 0$.

Let $\hat{W}_r \in \mathbb{R}^{1 \times o}$, $\hat{W}_a \in \mathbb{R}^{1 \times l}$, and $\hat{W}_v \in \mathbb{R}^{1 \times o}$ be the estimated weights, and $\sigma_1(\cdot) \in \mathbb{R}^l$ and $\sigma_2(\cdot) \in \mathbb{R}^o$ be the exponential activation functions, a graph neural network (GNN) is employed to construct the approximated value functions $\hat{V}_i$:

$$\hat{h}_j(e_{\mathcal{N}_j}^h) = \frac{1}{|\mathcal{N}_j|} \sum_{k \in \mathcal{N}_j} a_{k,j} \hat{W}_r \sigma_2(e_{k,j}^h), \quad (14)$$

$$\hat{V}_i(E(t)) = \max \left\{ \sum_{j \in \mathcal{N}_i} a_{i,j} \hat{W}_a \sigma_1(\hat{h}_j(e_{\mathcal{N}_j}^h(t))), \right. \\ \left. \max_{t \in [t, t+\tau]} \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}(t)) - T_\gamma)^{-1} \right\} \right\}, \quad (15)$$

where $e_{\mathcal{N}_j}^h = [\cdots, e_j^T, \cdots]^T : j \in \mathcal{N}_i$, and denoting $\hat{\mathbb{V}}_{i,j} \subseteq \mathbb{V}_{i,j}$ as the approximated viability kernel, $\hat{\gamma}(e_{i,j}) = \hat{W}_v \sigma_2(e_{i,j}) \in [0, \infty)$, if $e_{i,j} \in \partial \hat{\mathbb{V}}_{i,j}$, and otherwise $\hat{\gamma}(e_{i,j}) = \infty$, if $e_{i,j} \in \hat{\mathbb{V}}_{i,j} \setminus \partial \hat{\mathbb{V}}_{i,j}$. Similarly, $\hat{W}_r \sigma_2(e_{k,j}^h) \in [0, \infty)$, if and only if $e_{k,j}^h \in \hat{\mathbb{R}}_{k,j} \subseteq \mathbb{R}_{k,j}$, where $\hat{\mathbb{R}}_{k,j}$ is the approximated capture basin. Note that $\hat{W}_r$, $\hat{W}_a$ and $\hat{W}_v$ are shared variables among the network.

III. FORMATION CONTROL

By employing dynamic programming technique in terms of solving the value function V_i for the multi-player reach-avoid game, the theorem based on the quasi-variational Hamilton-Jacobi-Bellman (HJB) inequality proposed in [27], [28] is considered, where the studies in those literatures only investigated two-player reach-avoid games. This section involves approximating V_i , i.e., estimating the weights of the GNN, and then the optimal control strategy to the differential graphical game is designed.

A. Value Function Approximation

To present the theorem, we need the following assumption.

Assumption 3: There exists a constant c such that $\forall t, \Delta t > 0$:

$$|H(E(t), p) - H(E(t + \Delta t), p)| < c(|E(t) - E(t + \Delta t)|), \quad (16a)$$

where

$$H(E, p) = \sum_{j \in \mathcal{N}_i} \sum_{k \in \mathcal{N}_j} \frac{\partial V_i(E)}{\partial e_{k,j}^h} \left(f(e_{k,j}^h) + g_+ \left(e_{k,j}^h \right) u_k^* + g_- \left(e_{k,j}^h \right) u_j^* \right) + |\mathcal{N}_i|. \quad (16b)$$

Note that Assumption 3 guarantees the existence of a viscosity solution to the HJB inequality.

Now, the first result is presented in the following theorem:

Theorem 1: The value function V_i is the viscosity solution to the quasi-variational Hamilton-Jacobi-Bellman (HJB) inequality:

$$0 = \min \left\{ V_i(E) - \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}) - T_\gamma)^{-1} \right\}, H(E, p) \right\}. \quad (17)$$

Proof: Defining a smooth function

$$\mathcal{V}_i(E(t)) = |\mathcal{N}_i| \Delta t + \inf_{u_i \in U} \max \left\{ V_i(E(t + \Delta t)), \max_{t \in [t, t + \Delta t]} \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}(t)) - T_\gamma)^{-1} \right\} \right\}, \quad (18)$$

where $\Delta t \in [0, \tau]$. Then, we have

$$\begin{aligned} \mathcal{V}_i(E(t)) &= |\mathcal{N}_i| \Delta t + \inf_{u_i \in U} \max \left\{ \right. \\ &\quad \left. + \sum_{j \in \mathcal{N}_i} \varpi \left(e_{i,j}^h(t + \Delta t), u_i, u_j^* \right), \right. \\ &\quad \left. \max_{t \in [t, t + \tau]} \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}(t)) - T_\gamma)^{-1} \right\} \right\} \\ &= V_i(E(t)), \end{aligned} \quad (19)$$

and let $\Delta t \rightarrow 0$, (19) implies that

$$V_i(E(t)) \geq \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}(t)) - T_\gamma)^{-1} \right\}. \quad (20)$$

Based on (19) and (20), proving Theorem 1 is equivalent to prove that, if $V_i - \mathcal{V}_i$ has a local maximum at E , then

$$0 \leq \min \left\{ V_i(E) - \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}) - T_\gamma)^{-1} \right\}, \mathcal{H}(E, p) \right\}, \quad (21)$$

and if $V_i - \mathcal{V}_i$ has a local minimum at E , then

$$0 \geq \min \left\{ V_i(E) - \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}) - T_\gamma)^{-1} \right\}, \mathcal{H}(E, p) \right\}, \quad (22)$$

where

$$\begin{aligned} \mathcal{H}(E, p) &= \sum_{j \in \mathcal{N}_i} \sum_{k \in \mathcal{N}_j} \frac{\partial \mathcal{V}_i(E)}{\partial e_{k,j}^h} \left(f(e_{k,j}^h) \right. \\ &\quad \left. + g_+ \left(e_{k,j}^h \right) u_k^* + g_- \left(e_{k,j}^h \right) u_j^* \right) + |\mathcal{N}_i|. \end{aligned} \quad (23)$$

When $V_i - \mathcal{V}_i$ has a local maximum at E , then

$$\begin{aligned} &V_i(E(t + \Delta t)) - \mathcal{V}_i(E(t + \Delta t)) \\ &\leq V_i(E(t)) - \mathcal{V}_i(E(t)) \\ &\Rightarrow V_i(E(t + \Delta t)) - V_i(E(t)) \\ &\leq \mathcal{V}_i(E(t + \Delta t)) - \mathcal{V}_i(E(t)), \end{aligned} \quad (24)$$

and given Definition 2, the following inequality

$$\begin{aligned} &\max_{t \in [t, t + \Delta t]} \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}(t)) - T_\gamma)^{-1} \right\} \\ &\leq \max_{t \in [t + \Delta t, t + \tau]} \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}(t)) - T_\gamma)^{-1} \right\}, \end{aligned} \quad (25)$$

holds true. Thus, (19) implies

$$\mathcal{V}_i(E(t + \Delta t)) - \mathcal{V}_i(E(t)) + |\mathcal{N}_i| \Delta t \geq 0, \quad (26)$$

and let $\Delta t \rightarrow 0$, we have $\mathcal{H} \geq 0$, and (20) holds true, and hereby (21) is proved.

When $V_i - \mathcal{V}_i$ has a local minimum at E , then (19) implies

$$\mathcal{V}_i(E(t + \Delta t)) - \mathcal{V}_i(E(t)) + |\mathcal{N}_i| \Delta t \leq 0, \quad (27)$$

and let $\Delta t \rightarrow 0$, we have $\mathcal{H} \leq 0$, and hereby (22) is proved. The proof is completed. ■

B. Cooperative Learning

The capture basin and the viability kernel characterize the domains of $\hat{W}_r$ and $\hat{W}_a$ (i.e. $\hat{\mathbb{R}}_{i,j}$ and $\partial \hat{\mathbb{V}}_{i,j}$). The cooperative learning algorithm adapts the weights $\hat{W}_r$ and $\hat{W}_a$ shared throughout the network in an on-line manner. Denoting W_r and W_a as the ideal weights of the GNN, and $\varepsilon_r(e_{k,j}^h)$ and $\varepsilon_a(e_{k,j}^h)$ as the function reconstruction errors. We have

$$\begin{aligned} V_i(E(t)) &= \max \left\{ \sum_{j \in \mathcal{N}_i} a_{i,j} \left(W_a \sigma_1 \left(h_j \left(e_{\mathcal{N}_j}^h(t) \right) \right) \right. \right. \\ &\quad \left. \left. + \varepsilon_a \left(h_j \left(e_{\mathcal{N}_j}^h(t) \right) \right) \right), \max_{t \in [t, t + \tau]} \max_{j \in \mathcal{N}_i} \left\{ \right. \right. \\ &\quad \left. \left. (\gamma(e_{i,j}(t)) - T_\gamma)^{-1} \right\} \right\}, \end{aligned} \quad (28)$$

where

$$h_j(e_{\mathcal{N}_j}^h) = \frac{1}{|\mathcal{N}_j|} \sum_{k \in \mathcal{N}_j} a_{k,j} \left(W_r \sigma_2 \left(e_{k,j}^h \right) + \varepsilon_r \left(e_{k,j}^h \right) \right) \quad (29)$$

Remark 3: Under Assumption 2, the graph is assumed to be switching, therefore, the value function V_i is time-varying. Given (17), the Bellman error can be written as

$$\begin{aligned} \delta_i &= \min \left\{ \Gamma_i(E), \Gamma_1 \left(e^h, u^* \right) \right. \\ &\quad \left. + \Gamma_2 \left(e_{\mathcal{N}_i}^h, u_{\mathcal{N}_i \setminus i}^*, u_i \right) + \Gamma_\varepsilon \left(e^h, u^* \right) \right\} \\ &= \min \left\{ \Gamma_i(E), \tilde{\delta}_i \left(e^h, u^*, u_i \right) \right\}, \end{aligned} \quad (30)$$

where

$$\Gamma_i(E) = \hat{V}_i(E) - \max_{j \in \mathcal{N}_i} \left\{ (\gamma(e_{i,j}) - T_\gamma)^{-1} \right\}, \quad (31)$$

$$\Gamma_1 = - \sum_{j \in \mathcal{N}_i} \sum_{k \in \mathcal{N}_j \setminus i} \frac{a_{i,j} a_{k,j}}{|\mathcal{N}_j|} \left(\tilde{W}_r \tilde{\omega}_{k,j}^{r*} + \tilde{W}_r \tilde{\omega}_{k,j}^{r*} \right) \quad (32a)$$

$$\Gamma_2 = - \sum_{j \in \mathcal{N}_i} \frac{a_{i,j}^2}{|\mathcal{N}_j|} \left(\tilde{W}_r \tilde{\omega}_{i,j}^{r*} + \tilde{W}_r \tilde{\omega}_{i,j}^{r*} + \tilde{W}_r \nabla \sigma_2 \left(e_{i,j}^h \right) \hat{W}_a \nabla \sigma_1 \left(h_j \left(e_{\mathcal{N}_j}^h \right) \right) \tilde{u}_i \right) \quad (32b)$$

$$\Gamma_\varepsilon = - \sum_{j \in \mathcal{N}_i} \sum_{k \in \mathcal{N}_j} \left(\tilde{W}_a \nabla \sigma_1 \left(h_j \left(e_{\mathcal{N}_j}^h \right) \right) \nabla \varepsilon_r \left(e_{k,j}^h \right) + \nabla \varepsilon_a \left(h_j \left(e_{\mathcal{N}_j}^h \right) \right) \tilde{W}_r \nabla \sigma_2 \left(e_{k,j}^h \right) + \nabla \varepsilon_a \left(h_j \left(e_{\mathcal{N}_j}^h \right) \right) \nabla \varepsilon_r \left(e_{k,j}^h \right) \right) F_{k,j}^*. \quad (32c)$$

$$u_i = \begin{cases} \arg \min_{u_i \in U} \left\{ \sum_{j \in \mathcal{N}_i} \frac{a_{i,j}^2}{|\mathcal{N}_j|} \hat{W}_a \nabla \sigma_1 \left(e_{i,j}^h \right) \tilde{W}_r \nabla \sigma_2 \left(e_{i,j}^h \right) g_+ \left(e_{i,j}^h \right) u_i \right\}, & \text{if } \forall j \in \mathcal{N}_i, e_{i,j} \in \hat{\mathbb{V}}_{i,j} \setminus \partial \hat{\mathbb{V}}_{i,j}, \\ \arg \max_{u_i \in U} \left\{ \tilde{W}_v \nabla \sigma \left(e_{i,j} \right) g_+ \left(e_{i,j} \right) u_i \right\}, & \text{if } \exists j \in \mathcal{N}_i, e_{i,j} \in \partial \hat{\mathbb{V}}_{i,j}. \end{cases} \quad (34a)$$

$$u_i = \begin{cases} \arg \min_{u_i \in U} \left\{ \sum_{j \in \mathcal{N}_i} \frac{a_{i,j}^2}{|\mathcal{N}_j|} \hat{W}_a \nabla \sigma_1 \left(e_{i,j}^h \right) \tilde{W}_r \nabla \sigma_2 \left(e_{i,j}^h \right) g_+ \left(e_{i,j}^h \right) u_i \right\}, & \text{if } \forall j \in \mathcal{N}_i, e_{i,j} \in \hat{\mathbb{V}}_{i,j} \setminus \partial \hat{\mathbb{V}}_{i,j}, \\ \arg \max_{u_i \in U} \left\{ \tilde{W}_v \nabla \sigma \left(e_{i,j} \right) g_+ \left(e_{i,j} \right) u_i \right\}, & \text{if } \exists j \in \mathcal{N}_i, e_{i,j} \in \partial \hat{\mathbb{V}}_{i,j}. \end{cases} \quad (34b)$$

and Γ_1 , Γ_2 and Γ_ε are written in (32), where

$$F_{i,j} = f \left(e_{i,j}^h \right) + g_+ \left(e_{i,j}^h \right) u_i + g_- \left(e_{i,j}^h \right) u_j^*, \quad (33a)$$

$$F_{i,j}^* = f \left(e_{i,j}^h \right) + g_+ \left(e_{i,j}^h \right) u_i^* + g_- \left(e_{i,j}^h \right) u_j^*, \quad (33b)$$

$$\tilde{\omega}_{i,j}^r = \nabla \sigma_2 \left(e_{i,j}^h \right) \hat{W}_a \nabla \sigma_1 \left(h_j \left(e_{\mathcal{N}_j}^h \right) \right) F_{i,j}, \quad (33c)$$

$$\tilde{\omega}_{i,j}^{r*} = \nabla \sigma_2 \left(e_{i,j}^h \right) \tilde{W}_a \nabla \sigma_1 \left(h_j \left(e_{\mathcal{N}_j}^h \right) \right) F_{i,j}^*. \quad (33d)$$

Then, the control strategies are designed in (34): when $\forall j \in \mathcal{N}_i$, $e_{i,j} \in \hat{\mathbb{V}}_{i,j} \setminus \partial \hat{\mathbb{V}}_{i,j}$, the control strategy designed in (34a) aims to guarantee the convergence of $e_{i,j}^h \in \hat{\mathbb{R}}_{i,j}$; when $\exists j \in \mathcal{N}_i$, $e_{i,j} \in \partial \hat{\mathbb{V}}_{i,j}$, the control strategy designed in (34b) aims to guarantee collision avoidance and connectivity, i.e., $\forall j \in \mathcal{N}_i$, $\forall t \in [0, T]$, $e_{i,j}(t) \in \hat{\mathbb{V}}_{i,j}$. However, when taking two independent control strategies, the optimum convergence cannot be achieved. In the sense, the optimal control strategy can only rely on (34a) without adopting (34b) by the following theorem:

Theorem 2: When $\forall j \in \mathcal{N}_i$, $\hat{\mathbb{R}}_{i,j}$ is adapted ideally by

$$\hat{\mathbb{R}}_{i,j} = \hat{\mathbb{R}}_{i,j} \setminus \left\{ e_{i,j}^h \mid \exists \tau > 0, e_{i,j}(t + \tau) \in \partial \hat{\mathbb{V}}_{i,j} \right\}, \quad (35)$$

with control policy (34), then $\forall e_{i,j}^h \in \hat{\mathbb{R}}_{i,j}$, $e_{i,j} \notin \partial \hat{\mathbb{V}}_{i,j}$, and only control policy (34a) is adopted.

Proof: When $\exists j \in \mathcal{N}_i$, $e_{i,j}(t) \in \partial \hat{\mathbb{V}}_{i,j}$, the control policy (34b) is adopted, which guarantees that $\forall \tau > 0$, $e_{i,j}(t + \tau) \in \partial \hat{\mathbb{V}}_{i,j}$, then (35) removes $\partial \hat{\mathbb{V}}_{i,j}$ for $e_{i,j} \in \hat{\mathbb{V}}_{i,j}$. When the control policy (34a) is adopted, $\forall j \in \mathcal{N}_i$, there may exist a set of states $e_{i,j}^h(t) \in \hat{\mathbb{R}}_{i,j}$, $\exists \tau > 0$, $e_{i,j}(t + \tau) \in \partial \hat{\mathbb{V}}_{i,j}$, and (35) remove those set of states. Thus, if $\forall j \in \mathcal{N}_i$, $\hat{\mathbb{R}}_{i,j}$ is adapted ideally by (35), then $\forall \tau > 0$, $e_{i,j}(t + \tau) \notin \partial \hat{\mathbb{V}}_{i,j}$, and therefore, only one control policy (34a) is adopted. ■

The initial parameters during the backward set propagations are verified in each iteration, which guarantee $\forall e_{i,j}^h \in \hat{\mathbb{R}}_{i,j}$, $\varpi \left(e_{i,j}^h, u_i, u_j \right) \in [0, \infty)$, and $\forall e_{i,j} \in \partial \hat{\mathbb{V}}_{i,j}$, $\gamma(e_{i,j}) = \infty$ with the initialized control strategy derived from Pontryagin Minimum and Maximum Principle. The proposed on-line learning needs to guarantee the Bellman error δ_i to be always converging and

ultimately bounded, such that the control strategy designed in (34) and (35) can always adapt to a Nash equilibrium. Precisely, the learning process only adapts $\hat{W}_a$ and $\hat{W}_r$, while $\hat{W}_v$ remains its initial value. If $\hat{W}_a$ and $\hat{W}_r$ are adapted ideally, the following theorem holds:

Theorem 3: If $\hat{W}_a$ and $\hat{W}_r$ are ideal, and u_i is designed as shown in (34), then $|\delta_i|$ is bounded by $|\Gamma_\varepsilon(e^h, u^*)|$ ultimately.

Proof: Provided that the initial configuration in Assumption 2, i.e., $\forall j \in \mathcal{N}_i$, $e_{i,j}(0) \in \mathbb{V}_{i,j}$ and the designed control u_i in (34b) implies

$$\begin{aligned} & \forall t \in [0, T], \quad \forall t \in [t, t + \tau], \quad e_{i,j}(t) \in \hat{\mathbb{V}}_{i,j} \subseteq \mathbb{V}_{i,j} \\ & \Rightarrow \max_{t \in [t, t + \tau]} \max_{j \in \mathcal{N}_i} \left\{ \left(\hat{\gamma}(e_{i,j}(t)) - T_\gamma \right)^{-1} \right\} \\ & \geq \max_{j \in \mathcal{N}_i} \left\{ \left(\gamma(e_{i,j}(t)) - T_\gamma \right)^{-1} \right\} \\ & \Rightarrow \Gamma_i(E) \geq 0. \end{aligned} \quad (36)$$

Assuming $W_a = \hat{W}_a$ and $W_r = \hat{W}_r$, if $\mathbb{R}_{i,j}$ is adapted ideally such that $\forall j \in \mathcal{N}_i$, $\forall e_{i,j}^h \in \mathbb{R}_{i,j}$, $e_{i,j} \notin \partial \hat{\mathbb{V}}_{i,j}$, then, the designed control u_i in (34a) guarantees that

$$\begin{aligned} & \forall t \in [0, T], \quad \forall t \in [t, t + \tau], \quad e_{i,j}^h(t) \in \mathbb{R}_{i,j}, \\ & e_{i,j}(t) \in \hat{\mathbb{V}}_{i,j} \setminus \partial \hat{\mathbb{V}}_{i,j} \subseteq \mathbb{V}_{i,j}, \\ & \Rightarrow \max_{t \in [t, t + \tau]} \max_{j \in \mathcal{N}_i} \left\{ \left(\hat{\gamma}(e_{i,j}(t)) - T_\gamma \right)^{-1} \right\} = 0 \\ & \Rightarrow \Gamma_i(E) \geq 0. \end{aligned} \quad (37)$$

Thus, the bound of δ_i is determined by $\bar{\delta}_i$. Assuming $\hat{W}_a$, $\hat{W}_r$, and $\mathbb{R}_{i,j}$ are ideal, only (34a) is employed, by substituting (34a) into $\bar{\delta}_i$, we have $\bar{\delta}_i = \Gamma_\varepsilon(e^h, u^*)$, therefore $|\delta_i| \leq |\Gamma_\varepsilon(e^h, u^*)|$. The proof is completed. ■

Theorem 3 shows the minimal bound of the approximation error of the value function that the learning process can achieve to the most extend. Despite that $\hat{W}_a$ and $\hat{W}_r$ cannot be always ideal, the proposed adaptive laws ensure that they can always converge to their ideal values with bounded convergence errors, and the following theorems hold:

Theorem 4: Denoting λ as the forgetting factor, and assuming that $\frac{\omega_a}{\sqrt{1 + \omega_a^T \Lambda_a \omega_a}}$ and $\frac{\omega_r}{\sqrt{1 + \omega_r^T \Lambda_r \omega_r}}$ are persistently exciting

$$\dot{\mathcal{V}}_a + \dot{\mathcal{V}}_r = - [\tilde{W}_a \quad \tilde{W}_r] \begin{bmatrix} \frac{\omega_a \omega_a^T}{1 + \omega_a^T \omega_a} + \lambda \Lambda_a^{-1} & \frac{2 \omega_a^* \omega_r^T}{1 + \omega_r \omega_r^T} \\ \frac{2 \omega_r^* \omega_a^T}{1 + \omega_a \omega_a^T} & \frac{\omega_r \omega_r^T}{1 + \omega_r^T \omega_r} + \lambda \Lambda_r^{-1} \end{bmatrix} \begin{bmatrix} \tilde{W}_a^T \\ \tilde{W}_r^T \end{bmatrix} - 2\varphi_a - 2\varphi_r + \frac{\Gamma_\varepsilon \tilde{W}_a \omega_a}{1 + \omega_a^T \omega_a} + \frac{\Gamma_\varepsilon \tilde{W}_r \omega_r}{1 + \omega_r^T \omega_r}. \quad (40)$$

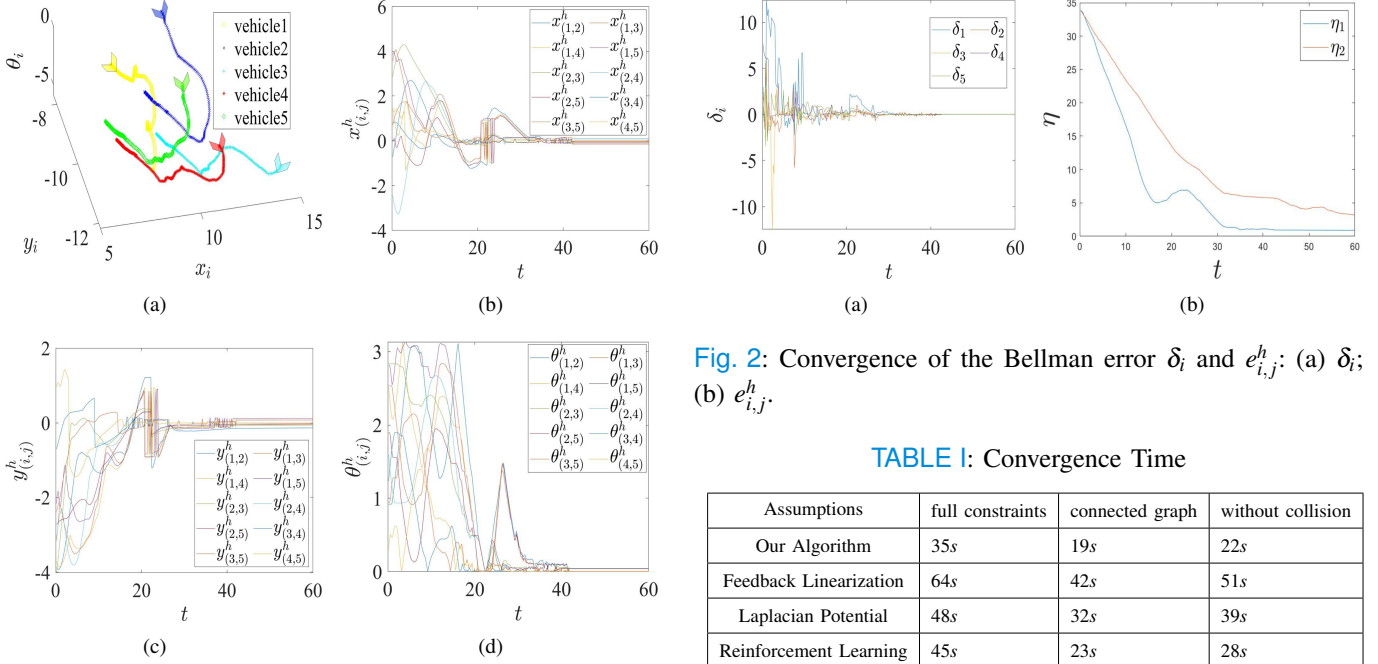


Fig. 1: Convergence of the states and the formation errors $e_{i,j}^h$: (a) the state trajectories in 3-D plots; (b) $x_{i,j}^h$; (c) $y_{i,j}^h$; (d) $\theta_{i,j}^h$.

[29], and $|\Gamma_\varepsilon|$ is bounded, the adaptive laws are designed as follows:

$$\begin{aligned} \dot{\hat{W}}_a = & - \sum_{j \in \mathcal{N}_i} \sum_{k \in \mathcal{N}_{j,i}} \frac{a_{i,j} a_{k,j}}{|\mathcal{N}_j|} \frac{\hat{W}_a \hat{\omega}_{k,j}^* \omega_a^T \Lambda_a}{1 + \omega_a^T \omega_a} \\ & - \sum_{j \in \mathcal{N}_i} \frac{a_{i,j}^2}{|\mathcal{N}_j|} \frac{\hat{W}_a \hat{\omega}_{i,j}^* \omega_a^T \Lambda_a}{1 + \omega_a^T \omega_a} - |\mathcal{N}_j| \frac{\omega_a^T \Lambda_a}{1 + \omega_a^T \omega_a} \end{aligned} \quad (38a)$$

$$\dot{\Lambda}_a = \lambda \Lambda_a - \frac{\Lambda_a \omega_a \omega_a^T \Lambda_a}{1 + \omega_a^T \omega_a}, \quad (38b)$$

and

$$\begin{aligned} \dot{\hat{W}}_r = & - \sum_{j \in \mathcal{N}_i} \sum_{k \in \mathcal{N}_{j,i}} \frac{a_{i,j} a_{k,j}}{|\mathcal{N}_j|} \frac{\hat{W}_r \hat{\omega}_{k,j}^* \omega_r^T \Lambda_r}{1 + \omega_r^T \omega_r} \\ & - \sum_{j \in \mathcal{N}_i} \frac{a_{i,j}^2}{|\mathcal{N}_j|} \frac{\hat{W}_r \hat{\omega}_{i,j}^* \omega_r^T \Lambda_r}{1 + \omega_r^T \omega_r} - |\mathcal{N}_j| \frac{\omega_r^T \Lambda_r}{1 + \omega_r^T \omega_r} \end{aligned} \quad (39a)$$

$$\dot{\Lambda}_r = \lambda \Lambda_r - \frac{\Lambda_r \omega_r \omega_r^T \Lambda_r}{1 + \omega_r^T \omega_r}, \quad (39b)$$

which guarantee the convergence of the estimation errors $\tilde{W}_a$ and $\tilde{W}_r$, where Λ_a and Λ_r are initialized as positive definite matrices. The control strategy designed in (34) always converges to a Nash equilibrium with the adaptive laws designed in (35), (38) and (39).

Fig. 2: Convergence of the Bellman error δ_i and $e_{i,j}^h$: (a) δ_i ; (b) $e_{i,j}^h$.

TABLE I: Convergence Time

Assumptions	full constraints	connected graph	without collision
Our Algorithm	35s	19s	22s
Feedback Linearization	64s	42s	51s
Laplacian Potential	48s	32s	39s
Reinforcement Learning	45s	23s	28s

Proof: Given (38) and (39), we have (40) by introducing Lyapunov-like functions $\mathcal{V}_a = \tilde{W}_a \Lambda_a^{-1} \tilde{W}_a^T$ and $\mathcal{V}_r = \tilde{W}_r \Lambda_r^{-1} \tilde{W}_r^T$. As $\|\tilde{u}_i\|$ has descending gradient with respect to $\|\tilde{W}_a\|$ and $\|\tilde{W}_r\|$, $|\varphi_a|$ and $|\varphi_r|$ have descending gradient with respect to $\|\tilde{W}_a\|$ and $\|\tilde{W}_r\|$. Then, (40) proves the convergence of $\tilde{W}_a$ and $\tilde{W}_r$ and shows they are ultimately uniformly bounded, if $|\Gamma_\varepsilon|$ is bounded. This implies that $\forall i \in [1, N]$, $|\delta_i|$ converges and can be ultimately uniformly bounded by the adaptive laws designed in (38) and (39). In the sense, all the constraints are satisfied, as $\forall i \in [1, N]$, $V_i \in [0, \infty)$, $|\delta_i| \in [0, \infty)$, otherwise $|\delta_i| = \infty$. Then, it can be concluded that the designed control law (34) always converges to a Nash equilibrium with the adaptive laws designed in (35), (38) and (39). The proof is completed. ■

IV. SIMULATION RESULTS

Fig. 1(a) illustrates the convergence of the states of five vehicles and the formation errors, respectively, where the terminal states shape a ‘star’, and the yellow vehicle has avoided a collision with the blue vehicle. Slight formation offsets are allowable after converging, as the target of $e_{i,j}^h$ is a set, i.e., $\mathbb{C}$, rather than 0. Fig. 1(b), 1(c) and 1(d) illustrates that the formation errors $e_{i,j}^h$ ultimately converge after $t = 35$.

Fig. 2(a) shows that all of δ_i remain finite values and converge to 0, which implies that the network system satisfies all the constraints with connectivity maintenance during the on-line learning process. Fig. 2(b) illustrates the comparison with the feedback linearisation based method with respect

to the convergence rate of states $x_{i,j}^h$ and $y_{i,j}^h$, i.e., $\eta = \sum_{i=1}^{N-1} \sum_{j=i+1}^N \left\| [x_{i,j}^h, y_{i,j}^h] \right\|$, in which η_1 is generated by the proposed method in this paper compared with η_2 generated by the feedback linearisation based method. It is observed that the convergence rate η_1 is apparently faster than η_2 . The statistical results of the convergence time generated by different algorithms with various constraints are demonstrated in TABLE I.

V. CONCLUSION AND FUTURE WORK

In this paper, we propose a novel framework for the formation control of constrained nonlinear network systems with multiple constraints based on viability theory. By linking the differential graphical game with viability theory, the proposed distributed control scheme is capable of aligning the desired displacement while satisfying all the constraints. By deploying adaptive GNN, the distributed control law enables the distributed control strategy to converge to a Nash equilibrium with connectivity maintenance despite of switching communication topology, where the designed scheme of the GNN adaptation mitigates the computational burden for the on-line cooperative reinforcement learning. The future work will be directed at distributed control for uncertain nonlinear network systems with disturbance rejection, coverage control for constrained multi-vehicle systems with arbitrary sensing scope of sensors, cooperative learning methods for traffic systems.

APPENDIX

A. Proof of Lemma 1

Proof: When (10) holds true, then, for $\forall t \in [0, T]$ and $\forall T \geq 0$, $\varpi(e_{i,j}^h(t), u_i, u_j) \in [0, \infty)$ and $\gamma(e_{i,j}(t)) \neq T_\gamma$. Provided that $e_{i,j}(0) \in \mathbb{V}_{i,j} \Leftrightarrow \gamma(e_{i,j}(0)) = \infty$, and $\gamma(e_{i,j}) = T_\gamma \Leftrightarrow e_{i,j} \in \partial V_{i,j}^c$, when $\exists u_i, u_j \in U$, $\forall T \geq 0$, $\forall t \in [0, T]$, $\forall \tau \in [0, T]$, $\gamma(e_{i,j}(t)) \neq T_\gamma \Rightarrow \exists u_i, u_j \in U$, $\forall t \in [0, \infty]$, $\gamma(e_{i,j}(t)) = \infty \Rightarrow \forall t \in [0, \infty]$, $e_{i,j}(t) \in \mathbb{V}_{i,j}$. When $\exists u_i, u_j \in U$, $\varpi(e_{i,j}^h(t), u_i, u_j) \in [0, \infty)$, then $\exists u_i, u_j \in U$, $\exists \tau \geq 0$, $e_{i,j}^h(t + \tau) \in \mathbb{C} \Rightarrow \forall t \in [0, T]$, $e_{i,j}^h(t) \in \mathbb{R}_{i,j}$, provided that $e_{i,j}(t) \in \mathbb{V}_{i,j} \subset \mathbb{K}$.

When $e_{i,j}^h \in \mathbb{R}_{i,j}$, then $\exists u_i, u_j \in U$, $\varpi(e_{i,j}^h, u_i, u_j) \in [0, \infty)$; when $e_{i,j} \in \mathbb{V}_{i,j}$, then $\exists u_i, u_j \in U$, $\gamma(e_{i,j}) = \infty$, and it implies that if $\forall t \in [0, T]$, $e_{i,j}(t) \in \mathbb{V}_{i,j}$, then $\exists u_i, u_j \in U$, $\forall t \in [0, T]$, $\max_{t \in [0, T]} \{(\gamma(e_{i,j}(t)) - T_\gamma)^{-1}\} = 0$. The proof is completed. ■

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